

2 **A title with some math: $x = 1$**

3 **Zhijie Zhao**

4 *Henan Normal University,*
5 *453007 Xinxiang, China*

6 *E-mail:* first@one.univ

7 ABSTRACT: Abstract...

8 ARXIV EPRINT: [1234.56789](https://arxiv.org/abs/1234.56789)

Process	Decay	Lower Mass Limit	Reference
$pp \rightarrow W'/Z'$	$V' \rightarrow WH/WZ/ff$	5.5 TeV	[1]
$pp \rightarrow Z'$	$Z' \rightarrow \ell^+\ell^-$	5.2 TeV	[2]
$pp \rightarrow W'$	$W' \rightarrow \ell\bar{\nu}_\ell$	5.7 TeV	[3]

Table 1. Current most stringent experimental limits on Z' and W' from LHC.

9 Contents

10	1 Constraints from LHC	1
11	1.1 Z' and W'	1
12	1.2 CP-odd Neutral Higgs Boson (A) and CP-even Neutral Higgs Boson (H)	2
13	1.3 $H^\pm$	2
14	1.4 $H^{\pm\pm}$	3
15	1.5 Vector-Like Lepton (VLL)	4
16	A Some title	4

17 1 Constraints from LHC

18 In this section, we summarize the current LHC limits on New Physics particles. The results
19 are based on data at a center-of-mass energy of 13 TeV, corresponding to an integrated
20 luminosity of 136-140 fb⁻¹.

21 1.1 Z' and W'

22 In the early 2026, CMS has published their combined statistical analysis of searches for
23 heavy vector boson resonances [1]. Both Drell–Yan (DY) and Vector Boson Fusion (VBF)
24 data are analyzed. For Z' , the decay channels $\ell^+\ell^-$, $q\bar{q}$, $t\bar{t}$, ZZ , Zh are considered. For
25 W' , the decay channels $\ell\bar{\nu}_\ell$ ($\ell = e, \mu, \tau$), $q\bar{q}'$, $t\bar{b}$, WZ , Wh are considered. This combined
26 analysis gives the most stringent constraints of Heavy Vector Triplet (HVT), where Z' and
27 W' are excluded with masses below 5.5 TeV.

28 For sequential standard model, the most stringent limits on Z' bosons come from high-
29 mass dilepton resonance searches, which set a lower mass limit on 5.2 TeV [2], while the
30 W' has a lower limit on 5.7 TeV, coming from $\ell\bar{\nu}_\ell$ channel [3].

31 We summarize these constraints in Table 1.

Process	Decay	Mass Range	Reference
$pp \rightarrow h$	$h \rightarrow AA \rightarrow 4\gamma$	0.1 – 1.5 GeV	[5]
$pp \rightarrow A$	$A \rightarrow \tau\tau$	20 – 90 GeV	[7]
$pp \rightarrow h$	$h \rightarrow AA \rightarrow \mu\mu\tau\tau$	3.6 – 21 GeV	[8]

Table 2. Current most stringent experimental limits on light A from LHC.

1.2 CP-odd Neutral Higgs Boson (A) and CP-even Neutral Higgs Boson (H)

LHC constraints on CP-odd neutral Higgs boson are evolved into two main regimes: light pseudoscalars (masses below the 125 GeV Higgs boson) and searches for heavy pseudoscalars (masses up to several TeV).

For particle A mass $m_A \leq 1$ GeV, the decay mode $A \rightarrow \gamma\gamma$ become important because it involves in photophilic scenarios [4]. CMS [5, 6] has studied the $pp \rightarrow h \rightarrow AA \rightarrow 4\gamma$ process. The result set a upper limit on the branching ratio $\mathcal{B}(h \rightarrow AA \rightarrow 4\gamma)$ down to $\mathcal{O}(10^{-3})$. When $m_A \geq 2m_\tau$, the decay $A \rightarrow \tau^+\tau^-$ becomes dominant. Ref. [7] reports a search for $pp \rightarrow A \rightarrow \tau^+\tau^-$ process. One τ decays to electron and neutrinos, while the other τ decays to muon and neutrinos. A mass range between 20 GeV and 90 GeV has been searched, and no significant excess is observed. Recently, Ref. [8] has searched for $h \rightarrow AA \rightarrow \mu^+\mu^-\tau^+\tau^-$ decay. Four decay modes $\tau\tau \rightarrow e\mu + \text{neutrinos}$, $\tau\tau \rightarrow e + \text{hadron(s)} + \text{neutrinos}$, $\tau\tau \rightarrow \mu + \text{hadron(s)} + \text{neutrinos}$, and $\tau\tau \rightarrow \text{hadrons} + \text{neutrinos}$ are considered. For $m_A \in (3.6, 21)$ GeV, the branching ratio is set to upper limit on $\mathcal{O}(10^{-4})$. Table 2 summarizes the searches for light A .

For the heavy A in TeV level, the Two-Higgs-Doublet Model (2HDM) is the most popular benchmark model in searching for A , as well as the CP-even Higgs boson H . In the 2HDM, the ratio of the vacuum expectation values (VEV) of two doublets, denoted as $\tan\beta$, is an important parameter. When $\tan\beta$ is large, say $10 \leq \tan\beta \leq 60$, the dominant decay modes of A/H are $A/H \rightarrow b\bar{b}$ or $\tau^+\tau^-$. Early searches [9, 10] for $pp \rightarrow A/H \rightarrow \tau^+\tau^-$ give strong constraints on m_A and m_H . The lower mass limits on m_A and m_H are set on around 2 TeV when $\tan\beta \geq 50$. For the low $\tan\beta$, the decay mode $A/H \rightarrow t\bar{t}$ becomes important since the coupling to top quarks is enhanced. Ref. [11] has searched for the $pp \rightarrow A/H \rightarrow t\bar{t}$ process, where one top quark decays leptonically and the other hadronically. Assuming $m_A = m_H$, the masses up to 1240 GeV are excluded for the lowest tested $\tan\beta = 0.4$. When $m_A > m_H + m_Z$, the decay $A \rightarrow ZH$ becomes possible and is dominant in a wide range of 2HDM parameter space. Ref. [12] has searched for $pp \rightarrow A \rightarrow ZH$ process, with $H \rightarrow t\bar{t}$. Z boson further decays to muons or electrons, while top quarks decay hadronically. The upper limits on $m_A = 2100$ GeV and $m_H = 2000$ GeV are obtained. The summary of constraints on heavy A and H are listed in table 3.

1.3 $H^\pm$

The searches for $H^\pm$ are categorized by its mass relative to the top quark: light ($m_{H^\pm} < m_t$) and heavy ($m_{H^\pm} > m_t$).

For the light $H^\pm$, top quark can decay to $H^\pm$ and b quark, and $H^\pm$ further decays to Standard Model particles. ATLAS has reported three decay modes: $H^+ \rightarrow \tau^+\nu_\tau$ [13],

Process	Decay	Upper Mass Limit	Reference
$pp \rightarrow A/H$	$A/H \rightarrow \tau^+\tau^-$	2000 GeV	[9, 10]
$pp \rightarrow A/H$	$A/H \rightarrow t\bar{t}$	1240 GeV	[11]
$pp \rightarrow A \rightarrow ZH$	$H \rightarrow t\bar{t}, Z \rightarrow \ell^+\ell^-$	$m_A = 2100$ GeV, $m_H = 2000$ GeV	[12]

Table 3. Current most stringent experimental limits on light A from LHC.

Process	Decay	Mass Range	Reference
$t \rightarrow H^\pm b$	$H^+ \rightarrow \tau^+\nu_\tau$	80 – 160 GeV	[13]
$t \rightarrow H^\pm b$	$H^\pm \rightarrow cs$	60 – 168 GeV	[14]
$t \rightarrow H^\pm b$	$H^\pm \rightarrow cb$	60 – 160 GeV	[15]
$pp \rightarrow tbH^\pm$	$H^\pm \rightarrow tb$	200-2000 GeV	[16]
$pp \rightarrow tbH^\pm$	$H^\pm \rightarrow Wh$	230-3000 GeV	[17]
$pp \rightarrow WbH^\pm b$	$H^\pm \rightarrow \tau\nu_\tau$	80-3000 GeV	[13]

Table 4. Current most stringent experimental limits on $H^\pm$ from LHC.

Process	Decay	Lower Mass Limit	Reference
$pp \rightarrow H^{++}H^{--}$	$H^{\pm\pm} \rightarrow \ell^\pm\ell^\pm$	1.08 TeV	[18]
$pp \rightarrow H^{\pm\pm}jj$	$H^{\pm\pm} \rightarrow W^\pm W^\pm$	1.5 TeV	[19]

Table 5. Current most stringent experimental limits on $H^{\pm\pm}$ from LHC.

$H^\pm \rightarrow cs$ [14] and $H^\pm \rightarrow cb$ [15]. For $H^\pm \rightarrow \tau\nu_\tau$, current limits on the branching fraction $\mathcal{B}(t \rightarrow H^\pm b) \times \mathcal{B}(H^\pm \rightarrow \tau\nu_\tau)$ are as low as 0.02% to 0.27% with $m_{H^\pm}$ from 80 GeV to 160 GeV. For $H^\pm \rightarrow cs$, upper limits on $\mathcal{B}(t \rightarrow H^\pm b)$ in the cs channel between 0.07% and 3.6%, assuming $\mathcal{B}(H^\pm \rightarrow cs) = 1$ and $m_{H^\pm} \in (60, 168)$ GeV. Similar to $H^\pm \rightarrow cs$, $H^\pm \rightarrow cb$ gives the upper limits on $\mathcal{B}(t \rightarrow H^\pm b)$ between 0.15% and 0.42%, where $m_{H^\pm} \in (60, 160)$ GeV.

For the heavy $H^\pm$, the $pp \rightarrow tbH^\pm$ has been widely studied since it is the dominant process of 2HDM model. The decay channels $H^\pm \rightarrow tb$ [16], $H^\pm \rightarrow Wh$ [17], and $H^\pm \rightarrow \tau\nu_\tau$ [13]. The mass range from 200 GeV to 3 TeV has been searched. The constraints for $H^\pm$ are summarized in Table. 4.

1.4 $H^{\pm\pm}$

At the LHC, doubly charged Higgs bosons can be produced via DY process or VBF process. Then they can decay to same signed leptons or vector bosons. Ref. [18] reports the search for DY production and the leptonic decay of $H^{\pm\pm}$. All combinations of lepton pairs have been considered, and they have the same branching ratio (1/6). This analysis gives a lower mass limit on 1.08 TeV. In another analysis [19], the VBF production and $H^{\pm\pm} \rightarrow W^\pm W^\pm$, where W further decays to lepton and neutrino, is studied. In this case, the mass range from 0.2 TeV to 1.5 TeV is excluded. The searches for doubly charged Higgs bosons are listed in table. 5.

Process	Decay	Lower Mass Limit	Reference
$pp \rightarrow e'^+ e'^-, e' N, NN$	$e' \rightarrow Z(H)e, N \rightarrow W\ell$	1220 GeV	[22]
$pp \rightarrow \mu'^+ \mu'^-, \mu' N, NN$	$\mu' \rightarrow Z(H)\mu, N \rightarrow W\ell$	1270 GeV	[22]
$pp \rightarrow \tau'^+ \tau'^-$	$\tau' \rightarrow \tau a_\tau \rightarrow \tau \gamma \gamma$	700 GeV	[23]
$pp \rightarrow L^+ L^-, LN, NN$	$L \rightarrow b\bar{b}\tau, t\bar{b}\bar{\nu}_\tau, N \rightarrow t\bar{t}\nu_\tau, t\bar{b}\tau^-$	910 GeV	[24]

Table 6. Current experimental limits on VLL from LHC.

1.5 Vector-Like Lepton (VLL)

VLLs are predicted in many extensions of the Standard Model, including composite Higgs models and theories addressing the muon $g-2$ anomaly. The production and decay modes for VLLs depend on the assumed $SU(2)$ representation [20, 21]. In the doublet scenario, two mass-degenerate VLLs at tree level, one electrically charged ($L^\pm$) and one electrically neutral (N), form an $SU(2)$ doublet (L, N). At the LHC, VLLs are predominantly produced in pairs via the electroweak interaction ($pp \rightarrow Z^*/\gamma^* \rightarrow L^+ L^-, pp \rightarrow Z^* \rightarrow N\bar{N}$ or $pp \rightarrow W^* \rightarrow L^+ \bar{N}$). The charged VLL decay modes are $L \rightarrow Z\ell$ and $L \rightarrow h\ell$. When $m_L \gg m_h$, the branching ratio reaches 50% for each modes. The neutral VLL N decays to lepton and W boson with 100% branching ratio. In the singlet scenario, only the charged VLL is present. It is also produced in pairs, $pp \rightarrow Z^*/\gamma^* \rightarrow L^+ L^-$. Its decay modes are $L \rightarrow W\nu, L \rightarrow Z\ell$ and $L \rightarrow h\ell$, with branching ratio 50%, 25% and 25%, respectively. Ref. [22] has searched for these processes and decay modes. Its result gives lower mass limits on 1220 GeV (1270 GeV) and 320 GeV (400 GeV) for vector-like electrons (muons) in the doublet and singlet scenarios, respectively.

In another paper [23], the pair production of vector-like tau (τ') has been considered. τ' decays to SM τ and a light long-lived pseudoscalar boson (a_τ). Since a_τ is long-lived particle, it leads to deposits in the muon system, which motivated experimentalists to search for this process. In the muon system, a_τ further decays to two photons, assuming it has mass below the $\tau^+ \tau^-$ threshold. In this analysis, the mass of a_τ is assumed to be $m_{a_\tau} = 2$ GeV. This analysis targets signals with at least one hadronically decaying τ and with at least one muon detector shower resulting from the a_τ decay in the CMS muon system. Its result excludes τ' masses up to 700 GeV.

Ref. [24] searches for VLL in the 4321 model. In this model, only one generation of VLLs is present in the model, and it is assumed to be fully mass degenerate. VLLs are produced through the pair production as mentioned above, and then they decay through a vector leptoquark U_1 as $L \rightarrow U_1 \bar{b}$ and $N \rightarrow \bar{U}_1 t$, followed by $U_1 \rightarrow t\bar{\nu}_\tau$ or $U_1 \rightarrow b\tau^+$. The branching ratios are $\mathcal{B}(L \rightarrow b\bar{b}\tau^+) = \mathcal{B}(L \rightarrow t\bar{b}\bar{\nu}_\tau) = 50\%$ and $\mathcal{B}(N \rightarrow t\bar{t}\nu_\tau) = \mathcal{B}(N \rightarrow t\bar{b}\tau^-) = 50\%$. Ref. [24] set a lower observed (expected) limit of 910 GeV on the VLL mass.

Table 6 summarizes the current LHC constraints on VLL.

A Some title

Please always give a title also for appendices.

Acknowledgments

This is the most common positions for acknowledgments. A macro is available to maintain the same layout and spelling of the heading.

Note added. This is also a good position for notes added after the paper has been written.

References

- [1] CMS collaboration, *Combination of searches for heavy vector boson resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV*, [2601.12583](#).
- [2] CMS collaboration, *Search for resonant and nonresonant new phenomena in high-mass dilepton final states at $\sqrt{s} = 13$ TeV*, *JHEP* **07** (2021) 208 [[2103.02708](#)].
- [3] CMS collaboration, *Search for new physics in the lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV*, *JHEP* **07** (2022) 067 [[2202.06075](#)].
- [4] D. Curtin et al., *Exotic decays of the 125 GeV Higgs boson*, *Phys. Rev. D* **90** (2014) 075004 [[1312.4992](#)].
- [5] CMS collaboration, *Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A} \rightarrow 4\gamma$ with events containing two merged diphotons in proton-proton collisions at $\sqrt{s} = 13$ TeV*, *Phys. Rev. Lett.* **131** (2023) 101801 [[2209.06197](#)].
- [6] CMS collaboration, *Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A}$ with $\mathcal{A} \rightarrow \gamma\gamma$ in events with a semi-merged topology in proton-proton collisions at $\sqrt{s} = 13$ TeV*, [2601.00183](#).
- [7] ATLAS collaboration, *Search for a light CP-odd Higgs boson decaying into a pair of τ -leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, *JHEP* **12** (2024) 126 [[2409.20381](#)].
- [8] CMS collaboration, *Search for $H \rightarrow a\bar{a} \rightarrow 2\mu 2\tau$ (boosted)*, .
- [9] ATLAS collaboration, *Search for heavy Higgs bosons decaying into two tau leptons with the ATLAS detector using pp collisions at $\sqrt{s} = 13$ TeV*, *Phys. Rev. Lett.* **125** (2020) 051801 [[2002.12223](#)].
- [10] CMS collaboration, *Searches for additional Higgs bosons and for vector leptoquarks in $\tau\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV*, *JHEP* **07** (2023) 073 [[2208.02717](#)].
- [11] ATLAS collaboration, *Search for heavy neutral Higgs bosons decaying into a top quark pair in 140 fb⁻¹ of proton-proton collision data at $\sqrt{s} = 13$ TeV with the ATLAS detector*, *JHEP* **08** (2024) 013 [[2404.18986](#)].
- [12] CMS collaboration, *Search for heavy neutral Higgs bosons A and H in the $t\bar{t}Z$ channel in proton-proton collisions at 13 TeV*, *Phys. Lett. B* **866** (2025) 139568 [[2412.00570](#)].
- [13] ATLAS collaboration, *Search for charged Higgs bosons produced in top-quark decays or in association with top quarks and decaying via $H^\pm \rightarrow \tau^\pm \nu \tau$ in 13 TeV pp collisions with the ATLAS detector*, *Phys. Rev. D* **111** (2025) 072006 [[2412.17584](#)].
- [14] ATLAS collaboration, *Search for a light charged Higgs boson in $t \rightarrow H^\pm b$ decays, with $H^\pm \rightarrow cs$, in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, *Eur. Phys. J. C* **85** (2025) 153 [[2407.10096](#)].

- [15] ATLAS collaboration, *Search for a light charged Higgs boson in $t \rightarrow H^{\pm}b$ decays, with $H^{\pm} \rightarrow cb$, in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, *JHEP* **09** (2023) 004 [[2302.11739](#)].
- [16] ATLAS collaboration, *Search for charged Higgs bosons decaying into a top quark and a bottom quark at $\sqrt{s} = 13$ TeV with the ATLAS detector*, *JHEP* **06** (2021) 145 [[2102.10076](#)].
- [17] ATLAS collaboration, *Search for a heavy charged Higgs boson decaying into a W boson and a Higgs boson in final states with leptons and b -jets in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector*, *JHEP* **02** (2025) 143 [[2411.03969](#)].
- [18] ATLAS collaboration, *Search for doubly charged Higgs boson production in multi-lepton final states using 139 fb^{-1} of proton-proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, *Eur. Phys. J. C* **83** (2023) 605 [[2211.07505](#)].
- [19] ATLAS collaboration, *Combination of searches for singly and doubly charged Higgs bosons produced via vector-boson fusion in proton-proton collisions at $s=13$ TeV with the ATLAS detector*, *Phys. Lett. B* **860** (2025) 139137 [[2407.10798](#)].
- [20] N. Kumar and S.P. Martin, *Vectorlike Leptons at the Large Hadron Collider*, *Phys. Rev. D* **92** (2015) 115018 [[1510.03456](#)].
- [21] P.N. Bhattiprolu and S.P. Martin, *Prospects for vectorlike leptons at future proton-proton colliders*, *Phys. Rev. D* **100** (2019) 015033 [[1905.00498](#)].
- [22] ATLAS collaboration, *Search for vector-like leptons coupling to first- and second-generation Standard Model leptons in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, *JHEP* **05** (2025) 075 [[2411.07143](#)].
- [23] CMS collaboration, *Search for vector-like leptons with long-lived particle decays in the CMS muon system in proton-proton collisions at $\sqrt{s} = 13$ TeV*, *JHEP* **08** (2025) 156 [[2503.16699](#)].
- [24] ATLAS collaboration, *Search for electroweak production of vector-like leptons in τ -lepton and b -jet final states in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, *Eur. Phys. J. C* **85** (2025) 1335 [[2503.22581](#)].